Document sympathique

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```r
### Library utiles
library(actuar)
```

```
##
## Attaching package: 'actuar'

## The following objects are masked from 'package:stats':
##
##     sd, var

## The following object is masked from 'package:grDevices':
##
##     cm
```

Exemple 1 : On $X \sim \text{PoComp}(\lambda = 2, \text{F}_\text{C})$ avec $C = 10B$ où $B \sim \text{Gamma}(3, 2)$. On discretise, $h = 0.1$ la v.a. C pour obtenir l'approximation $\tilde{X}$. Alors on constate que

$$F_C(x) = \Pr(C \leq x) = \Pr(10B \leq x) = \Pr(B \leq x/10) = F_B(x/10).$$

Donc,

```r
nfft <- 2^11
lam <- 2
a <- 3
b <- 2
h <- 0.1
k <- 0.95
x <- (0:(nfft - 1))*h # on ajuste le domaine de C en fonction des informations
fc <- pgamma(x/10, a, b) - pgamma((x-h)/10, a, b)

#Vérif
sum(fc)
```

```
## [1] 1
```

```r
espC_test <- 10*a/b
espC <- sum(fc * x)
cbind(espC, espC_test) # Génial
```

```
##       espC espC_test
## [1,] 15.05        15
```

```r
fxt <- exp(lam*(fft(fc) - 1))
fx <- Re(fft(fxt, TRUE))/nfft
Fx <- cumsum(fx)
#Vérif
sum(fc)
```

```
## [1] 1
```

```r
espX_test <- espC_test*lam
espX <- sum(fx * x)
cbind(espX, espX_test) # Génial
```

```
##          espX espX_test
## [1,] 30.09868        30
```

```r
varX <- sum(x^2 * fx) - espX^2

VaRX <- x[min(which(Fx >= k))]
TVaRX <- VaRX + 1/(1 - k)*sum(pmax(x - VaRX, 0)*fx)
```

Exemple 2 : On $X = \min(B, 100)$ où $B \sim \text{Gamma}(3, 2)$. On discretise, $h = 0.5$ la v.a. B pour obtenir l'approximation $\tilde{X}$. Alors on constate que

$$\Pr(\tilde{X} = x) = \begin{cases} \Pr(\tilde{B} = x), & x < 50 \\ \Pr(\tilde{B} > 49), & x \geq 50 \end{cases}$$

Donc,

```r
a <- 3
b <- 2
h <- 0.1
k <- 0.90
n <- 50
x <- (0:n)*h # on ajute le domaine de C en fonction des informations
fx <- pgamma(x[-(n + 1)], a, b) - pgamma((x[-(n + 1)]-h), a, b)
fx <- c(fx, pgamma((n-1)*h, a, b, low = FALSE))
Fx <- cumsum(fx)

sum(fx)
```

```
## [1] 1
```

```r
espX <- sum(x * fx)
varX <- sum(x^2 * fx) - espX^2

VaRX <- x[min(which(Fx >= k))]
TVaRX <- VaRX + 1/(1 - k)*sum(pmax(x - VaRX, 0)*fx)
```